Learning Surface Ozone From Satellite Columns (LESO): A Regional Daily Estimation Framework for Surface Ozone Monitoring in China

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***Abstract*— Continuously monitoring surface ozone (O_3) spatial distribution and forecasting its variations are beneficial to improving air quality and ensuring public health in China, although achieving this goal faces challenges from currently available observations and retrieval techniques. Hence, we introduce a coupled surface O_3 estimation framework (LESO) to address these challenges by integrating ground-level observing networks and satellite remote sensing. LESO features easy-to-use deep learning algorithms, independence on chemical transportation models (CTMs), and consistent performance using data from different satellites. LESO includes a deep forest 21 (DF21) model to interpolate O_3 concentration by learning spatial patterns and a long short-term memory (LSTM) model to forecast O_3 concentration by learning data from the past. We used sites of city-level *in situ* networks as the control sites to manifest short-distance O_3 transportation. Satellite-based observations of O_3 precursor indicators were incorporated to capture O_3 photochemical reactions. DF21 explained a larger fraction of O_3 variability (90%) with a mean bias error (MBE) of smaller than 1 $\mu g/m^3$. We also investigated the impact of the number of training sites on the DF21 performance, which suggested that five training sites could ensure a good DF21 performance for the most areas ($R^2 > 0.85$ and bias <2 $\mu g/m^3$). The forecast O_3 concentration via LSTM showed a good and stable agreement ($R^2 \approx 0.85$ and bias <5 $\mu g/m^3$) with ground-based measurements for 8-, 24-, 28-, and 72-h time periods, respectively. Overall, LESO aims to bring convenient functionality and reliable surface O_3 estimates for broad users.**

***Index Terms*— Air pollution, data assimilation, deep learning, remote sensing.**

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I. Introduction

COORDINATED control of fine particular matters ($PM_{2.5}$) and Ozone (O_3) marked a milestone for understanding and mitigating air pollution in China [1], [2]. In contrast to the decreased $PM_{2.5}$ in recent years, increasing levels of O_3 (by 3.4 $\mu g/m^3$/year) were seen in China [3], [4]. For over a decade, researchers have reported extensive $PM_{2.5}$-related study advances in its distribution, variation, and source apportionment [5]. Because its large-scale threats to public health were identified later, the studies focusing on O_3 are relatively insufficient compared to those of $PM_{2.5}$ [6]. The nationwide joint control of $PM_{2.5}$ and O_3 thereby can benefit from large-scale and fine spatiotemporal O_3 monitoring.

Surface O_3 is a secondary gaseous pollutant generated through photochemical reactions; its formation pathways depends on the ratio of its precursors: nitrogen oxides ($NO_x = NO + NO_2$) and volatile organic compounds (VOCs) [7]. Chemical transportation models (CTMs)-based methods are the mainstream approaches for estimating large-scale ground-level O_3 pollution [8], [9]. By incorporating photochemical reactions, CTMs are theoretically sound but the capabilities in explaining the concentration and distribution of O_3 over complicated terrains are largely limited [10]. Estimating surface O_3 by assimilating satellite remotely sensed O_3 columns, reanalyzed meteorology, and *in situ* networks has become a promising alternative [11].

Recently, a few new generation hyperspectral instruments have been launched and provide high-resolution and high-precision measurements of trace gases including ozone. TROPOspheric monitoring instrument (TROPOMI), a successor to ozone monitoring instrument (OMI), on board the Copernicus Sentinel-5 Precursor satellite measures total and tropospheric columns of O_3 with unprecedented spatial resolution, paving the way for reliably inferring ozone concentrations from space. However, estimating surface ozone directly from satellite measurements remains challenging because stratospheric ozone accounts for 90% of total ozone. Machine

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learning techniques have already been considered for surface O_3 estimation: 1) deep neural networks (DNNs) exhibited improved ground-level O_3 estimation accuracy [12]; however, for spatial estimation, many DNNs approaches require emission inventories, and the results were only validated in a small-sized region [12]; and 2) decision-tree-based machine-learning approaches are user-friendly compared to DNNs (e.g., simpler model structures and less dependence on high-performance computers), but their accuracy is lower than that of DNNs ($R^2 = 0.79$) even with a large amount of parameters [13]. For forecast, the recurrent neural network (RNN, a branch of DNNs designed for processing sequential tasks) collaborated with CTMs exhibited strong predictive accuracy for local-scale and/or near future (i.e., 24 h) O_3 forecasting tasks [14], [15]. Accordingly, a validation of larger scale and/or longer time span prediction is needed.

In this study, we propose an innovative algorithm framework for estimating spatial continuously O_3 (referred to as O_3 interpolation) and predicting future O_3 (referred to as O_3 forecast hereafter) based on the newly available TROPOMI measurements. The so-called Learning Surface O_3 from satellite columns (LESO) includes two major components: 1) a deep-forest (deep-layered yet decision-tree-based algorithm) for interpolation and 2) an improved RNN for forecasting. LESO is originated from a surface O_3 estimation study in a mountainous city in southwest China [16]. In addition to TROPOMI, LESO is also applied to OMI O_3 columns from 2014 to 2020 to generate a longer time series and to inter-compare results derived from two satellites. In Section II, we describe the architecture of LESO, the details and advantages of the two major components, the workflow of LESO implementation, and the data and validation indicators we used. In Section III, we analyze the performance of LESO in interpolation and forecasting, respectively, and discuss the superiorities of LESO to other benchmark algorithms. In Section IV, we sum up the recommended conditions for applying LESO in terms of bullet points for the convenience of other researchers. We validated the LESO approach in three highly polluted regions in China: Jing-Jin-Ji (JJJ), Yangtze River Delta (YRD), and Pearl River Delta (PRD). In addition to assessing the performance of LESO and analyzing its regional difference, we discuss in practice a minimum number of *in situ* sites are required for training a reliable LESO model and if the performance of LESO significantly degrades as the forecast time span increases from 8 to 72 h.

II. Methodology

A. LESO Model Architecture and Model Performance Evaluation Metrics

Fig. 1 shows the diagram of the LESO model for O_3 interpolation and forecast. The implementation workflow includes a dataset preparation part, two parallel parts, i.e., the DF-21-based interpolation and the LSTM-based forecast. The used deep-forest (DF21) can reconcile the conflicts in O_3 interpolation between DNNs-based and regular decision-tree-based approaches [17]. The used RNN (LSTM, long short-term memory) solved the gradient vanishing/exploding

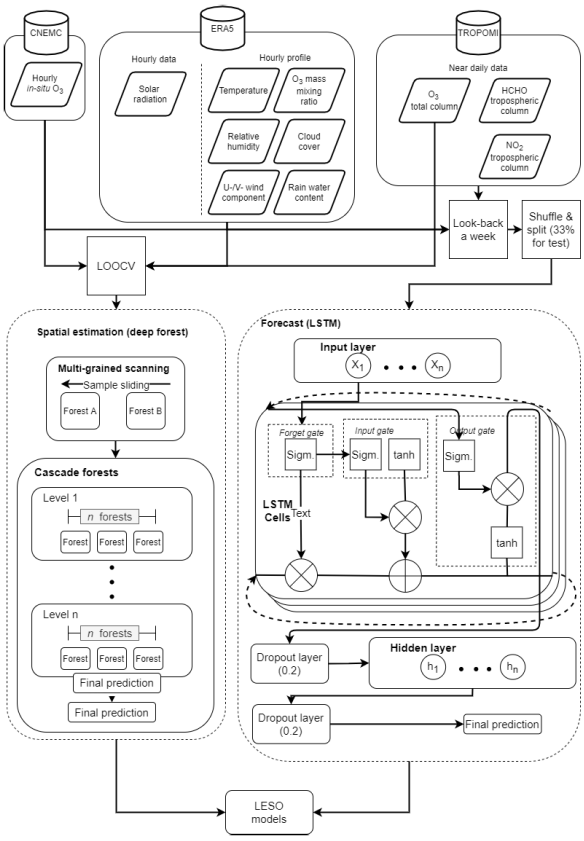


Fig. 1. Diagram of LESO, including data pre-processing, training/testing set preparation, and structure of interpolation and forecast sub-models.

issues in many other RNNs by introducing gates to keep or forget certain information [18].

1) Dataset Preparation: To create datasets for training/validating the LESO model, the gridded satellite data and reanalyzed meteorological parameters were interpolated into *in situ* sites. For interpolation, we adopted the leave-one-out cross-validation (LOOCV) [19] as the validation strategy, which means every site was used as the test set once while being part of the training set in the rest of the validations [20]. By this means, we can fully test the DF21 model fully for the dataset and reduce biases caused by validation strategies. For forecast, we separately predicted future O_3 concentration in 8, 24, 48, and 72 h based on data from one week before. In this case, the dataset was created by looking back to last week, and the conventional shuffle & split strategy [19] (test ratio 33%) was adopted for validation, i.e., one-third of randomly selected time-point data served as the test set and the others were the training set. The eXtreme Gradient Boosting (Xgboost) was adopted as the benchmark [21].

2) DF21 and O_3 Interpolation: Multigrained scanning and cascade forests are the two major components of deep forest DF21. Multigrained scanning is a sliding window feature generator, which is alike to the "convolution kernel" in convolutional neural networks. In this way, DF21 can generate more abundant feature relationships to facilitate model training. The cascade forests include layers of parallel forests; the individual forest can be a certain ensemble algorithm (e.g., random

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forest [22] and Xgboost). Here, DF21 is an ensemble of Xgboost. By stacking multiple layers, a deep forest is thereby established. The number of layers is adaptively regulated by comparing the cross-validated accuracy of current layer with previous one. The cascading process will stop if no accuracy improvement is seen [17].

DF21 features four advantages in comparison with DNNs and other decision-tree-based algorithms [17]: 1) DF21 usually gives a better prediction accuracy than that by other decision-tree-based algorithms; 2) although DF21 is a deep learning-based algorithm, it does not require users to determine model architecture, reducing user-related model uncertainty; 3) DF21 is easy-to-use with fewer hyper-parameters compared to DNNs; and 4) DF21 is capable of handling large-scale data and produces great performance on small-scale data. This is a major advantage compared to DNNs, because the data scale for interpolation task is much smaller than the data scale of classic deep learning tasks (e.g., computer vision) [17].

3) LSTM and O_3 Forecast: The reason of using LSTM [18] is due to its strong performance and robustness in handling time-series. Simply put, LSTM cells have "memory" as they store information from previous time-steps [18]. Each LSTM cell has an input gate, a forget gate, and an output gate. The function of the forget gate is to determine if the information of the previous cell is retained. Passing the information (of current input and previous hidden state) through the sigmoid activation function ($\in[0, 1]$), the information will be retained if the output value is close to one or forget otherwise. Here, we attached two dropout layers in consideration of model robustness [23]. The dropout layers will randomly drop a certain percentage (20% here) of nodes (i.e., the basic element of a neural network). The shuffle & split and dropout techniques are introduced to avoid potential overfitting issues. A fully connected layer locates between the two dropout layers, which is responsible for mapping the 64-node output vector from LSTM to a 16-node vector; and it will be finally mapped to one final prediction. The activation function for the fully connected layer is the rectified linear unit (ReLU) [24].

4) Statistical Measures: We used four statistical measures to assess model performance by comparing with the corresponding measurements in the test set. The determination coefficient (R^2) assesses the model accuracy (model explained variance in measured *in situ* O_3), in the following equation:

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(\mathrm{O_{3site}}_{(i)} - \mathrm{O_{3est}}_{(i)})^2}{\sum_{i=1}^{N}(\mathrm{O_{3site}}_{(i)} - \overline{\mathrm{O_{3site}}})^2} \tag{1}$$

where $\mathrm{O_{3\ est(i)}}$ is model estimated O_3 at site and $\mathrm{O_{3site}}_{(i)}$ is *in situ* measured O_3.

The linear regression slope (referred to as slope hereafter) assesses if over-/under-estimation exists in the models. The root mean squared error (RMSE) describes the model in the following equation:

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{N}(\mathrm{O_{3site}}_{(i)} - \mathrm{O_{3est}}_{(i)})^2}{N}}. \tag{2}$$

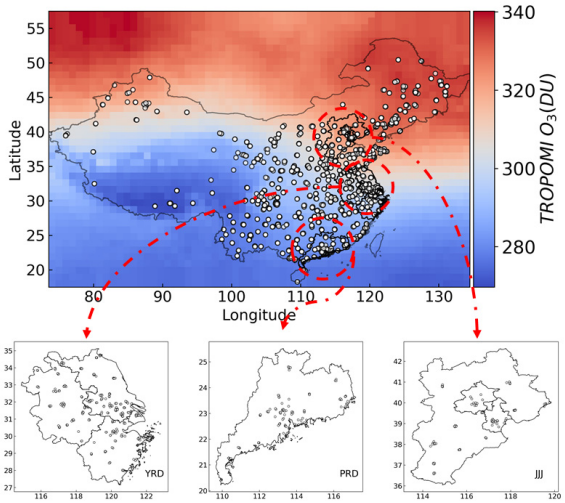


Fig. 2. Study areas JJJ, YRD, and PRD. The base map is the averaged summertime TROPOMI O_3 total column (2019 to 2020). White dots indicate the distribution of *in situ* sites.

The mean bias error (MBE) describes the bias between model predicts and measured O_3, in the following equation:

$$\mathrm{MBE} = \frac{\sum_{i=1}^{N}\mathrm{O_{3est}}_{(i)} - \sum_{i=1}^{N}\mathrm{O_{3site}}_{(i)}}{N}. \tag{3}$$

B. Study Area and Data

1) Study Areas: JJJ, YRD, and PRD are the regions exposed to severe O_3 pollution in China [6]. Fig. 2 demonstrates averaged summertime O_3 total columns over these three regions and the distribution of CNEMC sites. O_3 total columns are higher in northern areas, while O_3 tropospheric columns are higher in southern areas (see Fig. S1). JJJ includes two municipalities (Beijing and Tianjin) and one province (Hebei); YRD includes one municipality (Shanghai) and three provinces (Jiangsu, Zhejiang, and Anhui); PRD in mainland China contains a part of Guangdong province. Considering the city scale, this study focuses on seven large cities: Beijing, Tianjin, Shanghai, Nanjing (provincial capital of Jiangsu), Hangzhou (provincial capital of Zhejiang), Guangzhou (provincial capital of Guangdong), and Shenzhen (a megacity in Guangdong).

2) Data: Datasets included in model establishment were straightforward: the newly launched satellite [Sentinel-5P (precursor)] O_3 total columns (sensor: TROPOMI) [25] *in situ* measurements from the China National Environmental Monitoring Center (CNEMC), Beijing, China, and the fifth-generation ECMWF atmospheric reanalysis (ERA5) [26]. In addition, instead of tropospheric columns, O_3 total columns were used to exclude the inclusion of CTMs. The results using TROPOMI were inter-compared with results using the OMI Level-3 total O_3 columns at a $0.25° \times 0.25°$ resolution data (2014–2020) as an extra validation [27], [28].

TROPOMI/Sentinel-5P (launched in October 2017) is a part of ESA's Copernicus Program for monitoring global air pollution; it is a successor of NASA's long-run OMI.

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TROPOMI extends the capabilities of its predecessors; it is a sun-spectrometer measures various trace gases (e.g., O_3, SO_2, and methane) and clouds/aerosols via electromagnetic spectra ranging from ultraviolet (UV) to short-wavelength infrared (SWIR). TROPOMI has a wide swath width of ≈2600 km and a unprecedented spatial resolution at 3.5 km × 7 km (3.5 km × 5.5 km since August 6, 2019) [29].

Here, we used TROPOMI Level-3 gridded near real-time trace gas columns (time range: 2019–2020) derived from Level-2 products from Google Earth engine (GEE) at a 0.1° × 0.1° resolution. Details of the Level-2 near real-time O_3 products can be seen in the supplementary. GEE stores multipetabyte satellite/geo-spatial datasets, and it provides both free data access and an online data process platform [30]. TROPOMI provides level-3 global O_3 total column gridded data since mid-September 2018. In addition to O_3 columns, we used tropospheric NO_2 and HCHO columns derived from TROPOMI to capture O_3 photochemical reactions for the forecast study. TROPOMI NO_2 columns were available since late June 2018 and HCHO columns were available since December 2018.

Established since 2013, CNEMC provides hourly measurements on criteria air pollutants ($PM_{2.5}$ and O_3 etc.) across China. We used hourly O_3 measurements from 2019 to 2020. For the seven sample cities (see Table S1), Beijing has 12 sites (with a mean O_3 concentration of 64.16 μg/m^3), Tianjin has 15 sites (with a mean O_3 concentration of 64.38 μg/m^3), Shanghai has ten sites (with a mean O_3 concentration of 68.47 μg/m^3), Nanjing has nine sites (with a mean O_3 concentration 65.76 μg/m^3), Hangzhou has 11 sites (with a mean O_3 concentration of 55.64 μg/m^3), Guangzhou has 11 sites (with a mean O_3 concentration of 54.54 μg/m^3), and Shenzhen has 11 sites (with a mean O_3 concentration of 60.58 μg/m^3).

ERA5 provides various hourly-reanalyzed meteorology [26]. We used eight meteorological parameters (2019–2020) that could relate to the formation, dispersion, and depletion of O_3 and/or that could affect remote-sensing detection. The eight parameters are solar radiation ($h\upsilon$), fraction of cloud cover (cc), ozone mass mixing ratio ($O_{3\ (prof.)}$), relative humidity (rh), rainwater content (rwc), temperature (T), and U-/V-component of wind (u/υ). The vertical profiles of the latter seven parameters were defined as functions of five pressure levels (200, 500, 700, 900, and 1000 hPa).

Using these eight parameters (and HCHO/NO_2 columns), we considered the O_3 related photochemical reactions and atmospheric dispersion/transportation processes, in the following equation:

$$\begin{aligned} O_3(\text{col.}) = {} & P(O_{(3\text{surf.})}, O_{3(\text{prof.})}, h\upsilon, \text{rh}, NO_2, \text{VOC}, \text{rwc}, \text{cc}) \\ & + D(O_{3(\text{surf.})}, O_{3(\text{prof.})}, u, \upsilon, T, \text{rwc}, \text{cc}) + \delta \end{aligned} \quad (4)$$

where $O_{3(\text{col.})}$ is O_3 column, $O_{3(\text{surf.})}$ is surface O_3, VOC is volatile organic compounds, P is a nonlinear function of ensemble O_3-related photochemical reactions, D is O_3-related deposition, dispersion and transportation functions, and δ is random error. As the formation and depletion of tropospheric O_3 is a dynamic equilibrium regulated by the ratio of NO_2 to HCHO that acting as a proxy of VOCs [7], [31], [32], (5) and (6)

$$NO_2 + O_2 \xleftrightarrow{h\upsilon(\lambda<420\ \text{nm})} O_3 + NO \quad (5)$$

$$NO + \text{VOC} \longrightarrow NO_2. \quad (6)$$

Therefore, the vertical distribution of O_3, and meteorology conditions exert impacts on satellite retrieved O_3 columns. In virtual of machine-learning's capability of solving nonlinear problems, the O_3 formation and dispersion functions and error term can be summed up as a nonlinear function (F) of surface O_3 and all other parameters (denoted as params), in the following equation:

$$O_{3(\text{col.})} = F(O_{3(\text{surf.})}, \text{params}). \quad (7)$$

Surface O_3 is an inverse function of satellite O_3 columns and all the other parameters, in the following equation:

$$O_{3(\text{surf.})} = F^{-1}(O_{3(\text{col.})}, \text{params}). \quad (8)$$

Considering the spatial distribution of surface O_3, O_3 concentration from surrounding sites $O_{3(\text{surr.})}$ can be informative for each site, therefore the formula becomes a nonlinear function (G), in the following equation:

$$O_{3(\text{surf.})} = G(F^{-1}(O_{3(\text{col.})}, \text{params}), O_{3(\text{surr.})}). \quad (9)$$

Our machine learning-based algorithm is trained to learn the relationship between the surface O_3 and the satellite-observed O_3 columns together with the HCHO/NO_2 columns and meteorological parameters, as indicated in the equation above.

III. Results and Discussion

A. Interpolation (Spatial Estimation) Performance

DF21 R^2 for the most sites were higher than 0.8 [see Fig. 3(a)] and higher than Xgboost R^2 in all the eight provinces/municipalities [see Fig. 3(b)]. The median DF21 R^2 of 368 sites was 0.93, and the first quantile (Q1) reached 0.88 while the third quantile (Q3) for Xgboost was 0.82 (for details, see Table S2). R^2 of both DF21 and Xgboost showed regional differences, the box width (i.e., interquartile range) in JJJ was much narrower, whilst YRD was seen with much wider boxes and larger variations among provinces/municipalities (see Fig. 3). In terms of slope, RMSE, and MBE, the RMSE showed significant negative correlations (p-values $\ll 0.05$) with the R^2 for both DF21 and Xgboost, but DF21's RMSE was generally smaller than Xgboost's by ≈5 μg/m^3 (see Fig. 3 and Table S2). For DF21, the medians were 0.94 (slope), 9.27 μg/m^3 (RMSE), and −0.10 μg/m^3 (MBE); for Xgboost, the medians were 0.67 (slope), 15.54 μg/m^3 (RMSE), and −0.48 μg/m^3 (MBE) (see Table S2). In addition, the slope of the DF21's Q1 was higher than that of the Xgboost's Q3, while the RMSE of the DF21's Q3 was smaller than that of the Xgboost's Q1 (see Table S2). No obvious regional differences were seen for slope, RMSE, and MBE (see Figs. 3 and S2).

The proposed DF21 showed overall promising interpolation performance, and it outperformed Xgboost in all three regions (JJJ, YRD, and PRD) (see Figs. 3 and S2, and Table S2).

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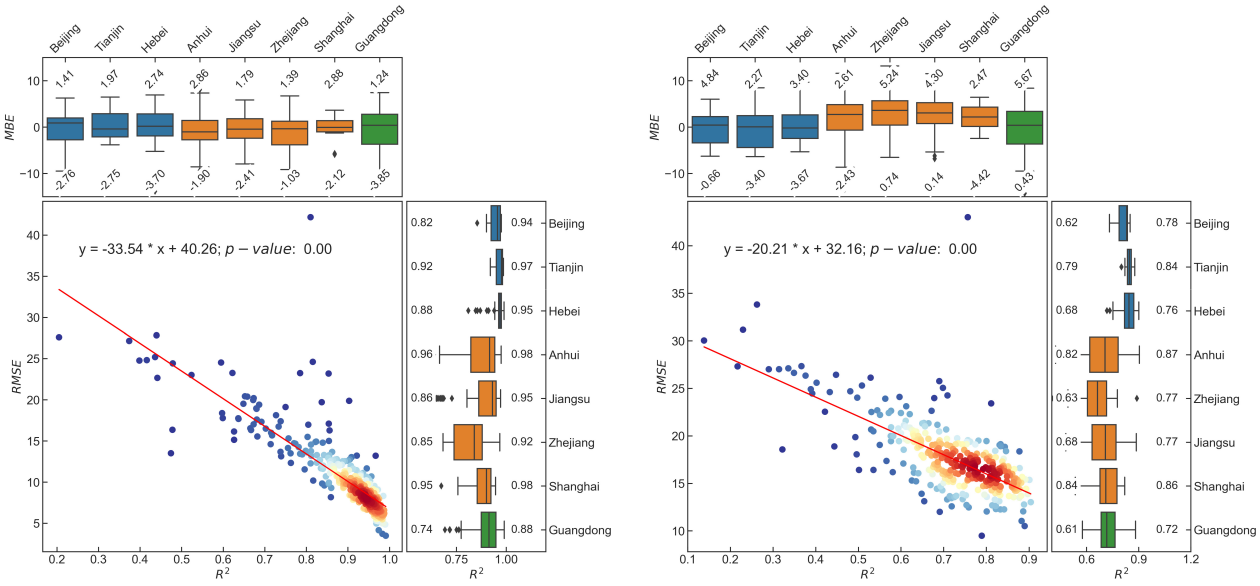


Fig. 3. Interpolation performance at 368 sites of 82 cities in eight provinces/municipalities of the three regions (JJJ, YRD, and PRD) for (a) DF21 and (b) Xgboost, respectively. Each panel (a) and (b) includes a density scatter plot of R^2 against RMSE (the redder the dots, the more concentrated distribution) and two boxplots. The right boxplot shows the statistical distribution of R^2 in the eight provinces/municipalities while the boxplot on the top shows the statistical distribution of MBE. Blues boxes stand provinces/municipalities for JJJ, orange boxes for YRD, and the green box for PRD. The first quartile (Q1) and the third quartile (Q3) values are separately given on both sides of the boxplots. The unit of RMSE and MBE are μg/m^3.

The superiority to Xgboost was anticipated, because its more complicated model architecture enables DF21 to fully extract and exploit the useful information even from a small dataset. DF21 produced a very high R^2 median of 0.93, as compared to those of Xgboost (≈0.77), the random forest (≈0.79) [13], and the deep-learning neural networks (≈0.90) [12]. This suggests that deep-learning methods could be considered as the mainstream surface O_3 estimation method in the future. DF21 might be preferable due to its user-friendly APIs and because it requires less efforts on tunning parameters and moderate computer hardware. In this study, the computation for interpolation was conducted on a laptop (16-GB RAM and Intel(R) Core (TM) i7-8665U CPU @ 1.90 GHz 2.11 GHz), significantly cheaper than the computation resources for DNNs used in previous studies, e.g., [33], [34]. Nevertheless, the weakness of DF21 lies on the long computational time. The time for running DF21 was at least double of that of Xgboost and was expected to be much longer than that of the GPU supported DNNs.

DF21 predicted O_3 time-series was consistent with CNEMC measurements, and this consistency was generally found all cities in Fig. S3. The seasonality was correlated with latitude: Beijing and Tianjin in North China showed a strong O_3 seasonality (high-levels of O_3 in summer and low-levels of O_3 in winter) (see Fig. S3); while Guangzhou and Shenzhen in South China were seen with a shifted O_3 seasonality (peaks seen in autumn) (see Fig. S3). The shaded areas of DF21 predicted O_3 concentrations, generally were coincided with CNEMC measurements (see Fig. S3). The occasionally narrower shaded areas suggested that DF21 slightly underestimated surface O_3, agreeing with the median slope of 0.94 (see Table S2).

According to the high R^2 from winter 2019 to autumn 2020 (red dots in Fig. 4), DF21 effectively captured O_3 variability throughout all the seasons. The highest O_3 concentrations were seen in the summer for North China, e.g., JJJ and the mid-north part of YRD; for South China, e.g., PRD and the south part of YRD, the highest O_3 concentrations were seen in the autumn (see Fig. 4). The same seasonal characteristics were also observed in the results using OMI columns (2014 to 2020) (see Fig. S3) and in the literature [35], [36]. In line with previous studies, increased O_3 concentrations were observed during the COVID-19 lockdown in spring 2020 [37], [38], but O_3 concentrations in autumn 2020 were lower than in autumn 2019, which could be attributed to the continuous but intermittent shutdown in all sectors of China (see Fig. 4).

The DF21 model has also been tested in other regions of mainland China, the averaged R^2 was higher than 0.8 and the averaged MBE in most cities was much smaller than 1 μg/m^3. We only present results of JJJ, YRD, and PRD because these three regions are the main urban agglomeration in China and are exposed to severe O_3 pollution [6].

B. Selection of Training Sites and Their Impacts on the Interpolation Performance

The number of training sites and the distance between the test site and training sites can have impacts on the interpolation performance. These impacts are considered to be critical, albeit not having been discussed previously.

DF21 R^2 and RMSE values appeared to be positively and negatively correlated with the number of training sites, respectively (see Fig. 5, and Tables I and S3). The

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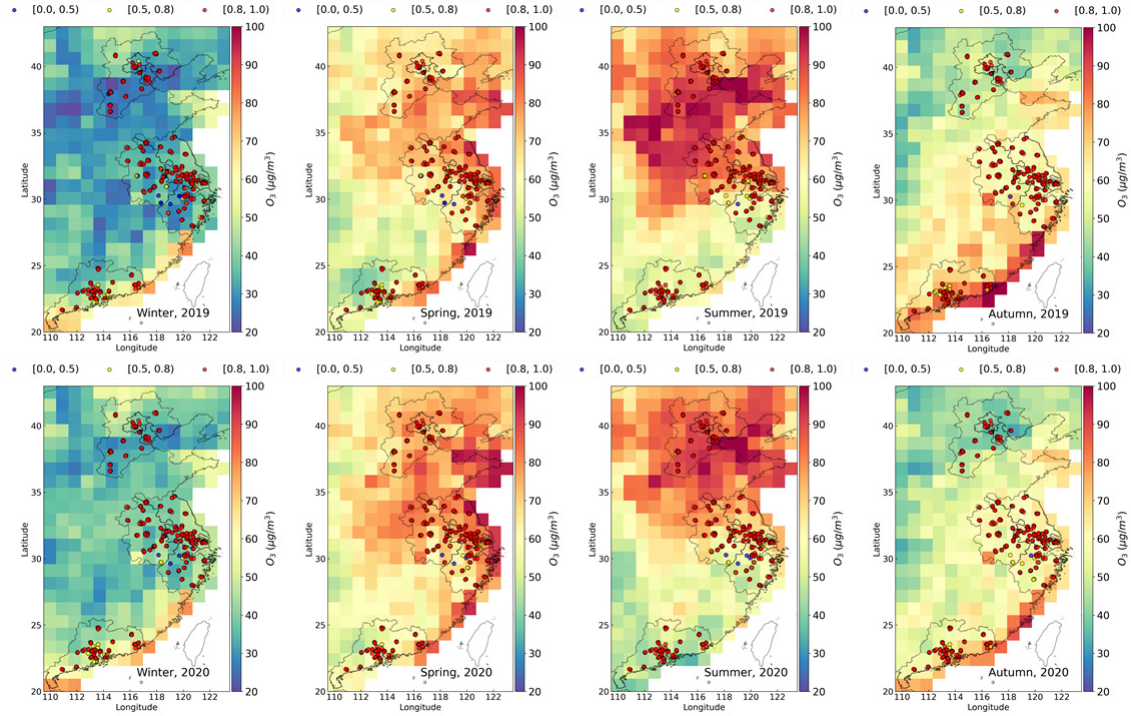


Fig. 4. Seasonal averages of DF21 predicts at a 1° × 1° spatial resolution (see Fig. S3 for the higher resolution version) in eastern China, including JJJ, YRD, and PRD; the dots are locations of CNEMC sites and the colors represent the validation R^2. Besides, time-series of CNEMC measured O_3 and DF21 predicts (the curves represent seven-daily averages for better visual effects and the shaded areas represent the standard deviation among sites) can also be found in Fig. S3 for both TROPOMI and OMI.

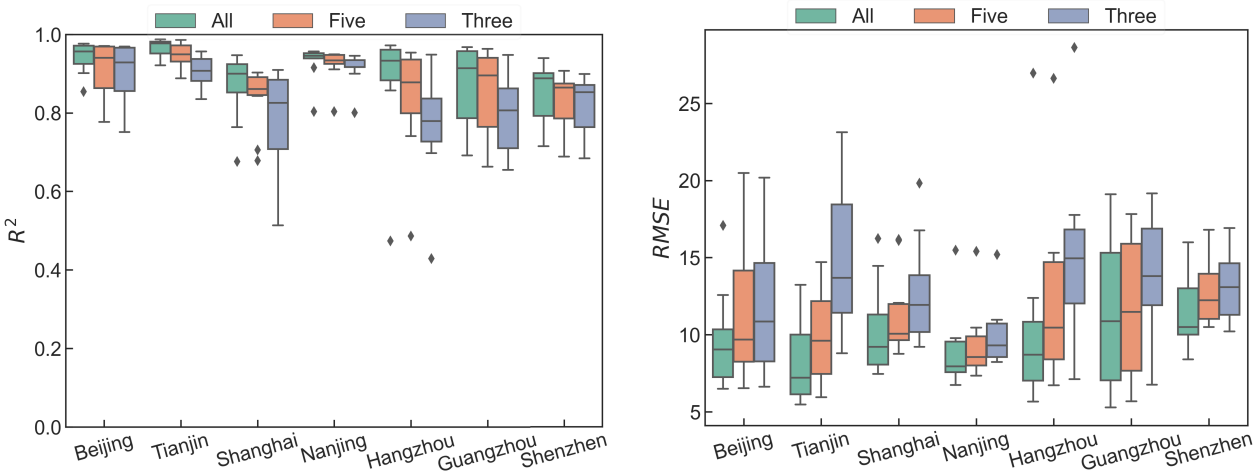


Fig. 5. Relationship between the DF21 results and the number of training sites (All: all available training sites; Five: randomly selected five sites; Three: randomly selected three sites). Box plot of (a) R^2 and (b) RMSE of the seven cities in JJJ, YRD, and PRD grouped by the number of training sites. Beijing, Tianjin, Shanghai, Nanjing, Hangzhou, Guangzhou, Shenzhen have 12, 15, 10, 9, 11, 11, and 11 sites in total, respectively.

R^2 median for all cities except Shanghai were around 0.9 with at least five training sites; the slope dropped slightly as the number of training sites declined; but no obvious difference was seen in MBE in accordance with the number of the training sites (see Table I). In this case, it can be inferred that at three training sites required for a favorable interpolation performance (e.g., $R^2 > 0.8$) and using five training sites can a promising performance (e.g., R^2 approx. 0.9). Nevertheless, there may be only one or two training sites in some cities. Besides, the locations of training sites should also be taken into consideration in practice.

Evidently, DF21 R^2 increased and RMSE decreased as the number of training sites increased, when the number of training sites was fewer than three [see Fig. 6(a)]. The improved interpolation performance became negligible when the training

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TABLE I

MEDIANS OF DF21 PERFORMANCE (IN STATISTICAL MEASURES) WITH DIFFERENT NUMBER OF TRAINING SITES. THE UNIT OF RMSE AND MBE ARE $\mu g/m^3$

	R^2	Slope	RMSE	MBE	R^2	Slope	RMSE	MBE	R^2	Slope	RMSE	MBE
	All				Five				Three			
Beijing	0.96	0.95	9.03	0.89	0.94	0.93	9.68	-0.81	0.93	0.93	10.84	0.04
Tianjin	0.98	0.97	7.21	-0.43	0.95	0.96	9.60	0.06	0.91	0.90	13.69	-4.01
Shanghai	0.90	0.89	9.21	-0.01	0.86	0.84	10.07	-1.56	0.83	0.78	11.92	-1.63
Nanjing	0.94	0.94	7.94	-0.57	0.93	0.94	8.55	0.27	0.92	0.92	9.31	-0.48
Hangzhou	0.93	0.90	8.70	-0.97	0.88	0.90	10.45	-0.90	0.78	0.81	14.96	-0.89
Guangzhou	0.91	0.92	10.86	-0.63	0.90	0.94	11.47	-0.45	0.81	0.81	13.80	-0.58
Shenzhen	0.89	0.89	10.49	0.37	0.86	0.86	12.23	2.00	0.85	0.86	13.07	1.68

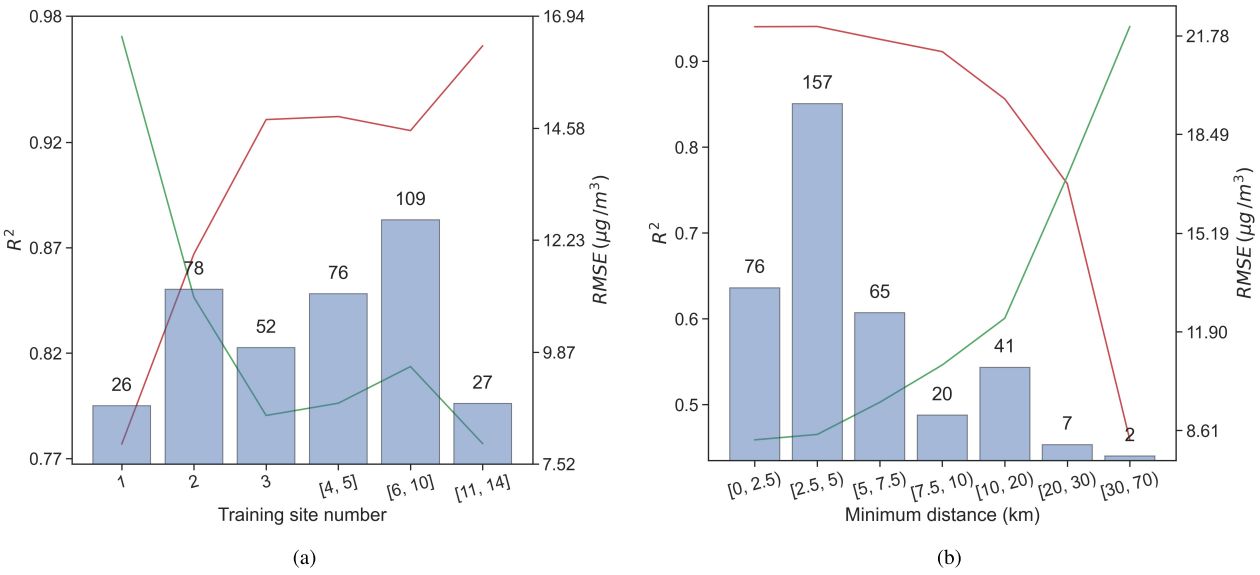


Fig. 6. Further analyses on the relationship between DF21 interpolation performance and the (a) number of training sites and the (b) minimum distances of training sites to the test site. The bars reflect the site numbers binned by their training site numbers. The red lines represent the R^2 of DF21 and the green lines represent the corresponding RMSE.

site number was between three and ten [see Fig. 6(a)]. Despite the large increase in the interpolation performance when the training sites were more than ten, it is not feasible to have that many training sites in any city of China [see Fig. 6(a)]. In fact, 49% of cities in this study have fewer than three CNEMC sites (82% of them are in YRD, 13% in PRD, and 5% in JJJ). In this regard, at least three to five training sites for one city would be highly beneficial to well-performed surface O_3 interpolations. By testing all sites in the three regions, DF21 with three to five training sites can ensure the performance of $R^2 > 0.8$ with 90% confidence and that of $R^2 > 0.9$ with 69% confidence.

The distribution of training sites could also have impacts on the interpolation performance. The minimum training site distance (i.e., distance between the test site and the closest training site, the same hereafter) was seen highly correlated with the interpolation performance [see Fig. 6(b)]. The closer the minimum training site distance, the better interpolation performance (higher R^2 and smaller RMSE). However, this does not mean all the sites should be co-located. For example, 21% sites were seen with <2.5 km, but the median of R^2 of sites with minimum distance <2.5 km is only 0.03 higher than of sites with minimum distance in [7.5, 10) km and only 0.08 higher than of sites with minimum distance in [10, 20) km [see Fig. 6(b)]. Besides, the spatial coverage would be very limited if all the sites are operated too closely. It can be suggested to plan the future sites distributed more evenly with a distance between 10 and 20 km, and the trade-off could be a decrease of 0.08 in R^2 and an increase of 4.06 $\mu g/m^3$ in RMSE.

C. Forecast (Temporal Estimation) Performance

LSTM forecast future O_3 with an encouraging accuracy ($R^2 \approx 0.87$), and the R^2 of PRD was higher those of YRD and JJJ; the mean RMSE for LSTM forecast O_3 was 18 $\mu g/m^3$ (see Fig. 7 and Table II). In comparison, the previous studies mainly focused on 8-h average/maximum daily O_3 forecast [14], [39], and the R^2 ranged from ~0.69 to ~0.8 [15], [39]. The performance of LSTM was reliable as the forecast time period increased (see Fig. 7, and Tables II and S4). This can be accounted for by: 1) a large amount of training data (ca. 2/3 of the whole two-year-long time series) when the forecast time period increases from 8 to 72 h; and 2) the LSTM forget gate that overcomes the

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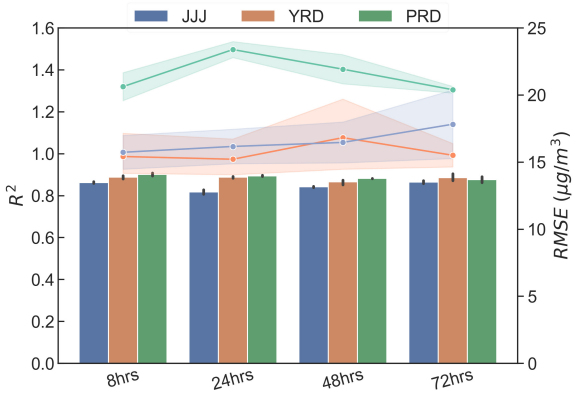


Fig. 7. O_3 forecast performance variations along the forecast time period, i.e., 8 h (8 h), 24 h (24 h), 48 h (48 h), and 72 h (72 h). The bar plot reflects the region-averaged variations in R^2 and the line plot represents the variations in RMSE. The colors identify the three regions.

gradient vanishing/exploding issues for long time series by disregarding a certain proportion of information in the back propagation [18]. The variations of R^2 was smaller than 0.05 (<5%). The variations of slope was lower than 0.06 (<6%). The variations of RMSE was lower than 3 μg/m^3 (<15%). The range of the variations of MBE was 1.10 μg/m^3 (<40%) for JJJ, and it was lower than 0.02 μg/m^3 (<2%) for YRD and PRD. Therefore, LSTM exhibited promising city-level O_3 forecast performance.

The time-series predicted by LSTM were in an overall good agreement with the CNEMC measurements (see Fig. S5). The LSTM results did not show any evident inter-city differences (see Fig. S5). In JJJ [see Figs. S5(a) and 7(b)], the comparison showed slightly larger disagreement around July 2019 from 24- to 72-h forecasting. In YRD [see Fig. S5(c), and Figs. 7(d) and 7(e)], the comparison showed a similar agreement between the predicted O_3 time-series and measured O_3 for different time periods. In Guangzhou of PRD, a slight disagreement was found in October 2019 for 8- and 24-h forecast [see Fig. S5(f)].

LSTM outperformed the benchmark model Xgboost (see Tables II, S4, and S5), the median R^2 for Xgboost was about 0.75 and the mean RMSE for Xgboost was 25 μg/m^3 (see Fig. S4 and Table S5). As compared to LSTM, MBE for Xgboost forecast O_3 was smaller (see Table S5). The performance of Xgboost was slightly worse than that of LSTM as the forecast time period increased (see Table S6). The variations of R^2 were lower than 0.06 (<8%). The variations of slope were lower than 0.08 (<14%). The variations of RMSE were lower than 3 μg/m^3 (<13%). The range of the variations of MBE was lower than 0.50 μg/m^3 (<39%), lower than 0.40 μg/m^3 (<212%), and lower than 1.20 μg/m^3 (<206%) for JJJ, YRD, and PRD, respectively. In terms of the reproduced time-series, the O_3 time-series predicted by Xgboost seemed to have large discrepancies compared to the CNEMC measurements (see Fig. S6). Xgboost missed to capture spiky peaks for longer forecast time periods especially for 72-h forecast (e.g., Fig. S6(a) row 3 and (g) row 4).

In addition to the seven high-level O_3 exposed big cities, we also tested the performance of LSTM in smaller cities, e.g., Huzhou, a city with three CNEMC sites in Zhejiang. In Huzhou, the median R^2 and RMSE of LSTM O_3 for time periods of 8, 24, 48, and 72 h was 0.88 ± 0.01 and 16.87 ± 0.56 μg/m^3. Therefore, using LSTM to forecast surface O_3 concentration is proven to be applicable for both large and small cities.

TABLE II

STATISTICAL MEASURE MEDIANS OF LSTM PERFORMANCE FOR 8-, 24-, 48-, AND 72-H O_3 FORECAST IN THE THREE REGIONS. THE UNIT OF RMSE AND MBE ARE μG/M^3

	R^2	Slope	RMSE	MBE
		JJJ		
8hrs	0.86	0.81	20.62	-3.04
24hrs	0.82	0.78	23.38	-2.89
48hrs	0.84	0.82	21.92	-3.12
72hrs	0.86	0.81	20.39	-3.99
		YRD		
8hrs	0.89	0.86	14.93	0.89
24hrs	0.89	0.90	14.85	0.89
48hrs	0.87	0.84	16.29	0.87
72hrs	0.88	0.85	15.49	0.88
		PRD		
8hrs	0.90	0.87	15.73	0.90
24hrs	0.89	0.91	16.16	0.89
48hrs	0.88	0.91	16.46	0.88
72hrs	0.88	0.91	17.82	0.88

D. Limitations of LSTM Applications and Analysis on Longer Forecast Periods

Despite the favorable forecast performance, some concerns remain on:

1) *The Complexity of Building and Applying an LSTM Model:* LSTM may require GPU for implementation and has more hyper-parameters for fine-tuning. LSTM's APIs are more obscure to nonexpert users, and the trained models are less portable in contrast to DF21 and other shallow-layered methods [40];
2) *Potential Challenges in Forecasting Over Longer Periods:* LSTM showed stable performance in forecasting O_3 from in 8 h to in three-day (see Fig. 7). Theoretically, LSTM may experience a turning point where its performance would degrade rapidly.

To pinpoint the limits of LSTM, e.g., in Nanjing, we applied LSTM in forecasting O_3 in one-week, two-week, one-month, three-month, and six-month. The R^2 ranged from 0.86 to 0.90, while the RMSE ranged from 15.17 to 17.56 μg/m^3. This infers the LSTM might be "exaggeratedly" robust in O_3 forecast. This unrealistic robustness could be related to the seasonality and diurnal patterns of O_3. Fig. S7(a) shows the monthly averaged diurnal patterns using the CNEMC measured O_3. Combined with Fig. 4, it can be seen that O_3 concentrations reflected remarkable seasonal and diurnal patterns. In this case, the LSTM can be employed to forecast the O_3 concentration within a year. However, at finer spatial scale (e.g., site-level) or in regions that might experience frequent large concentration variations, LSTM might be less robust.

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Furthermore, the capability of LSTM in temporal extrapolation was tested using data from 2014 to 2017 to predict future O_3 in 2018–2020 [see Fig. S7(b)]. LSTM successfully captured the long-term trend and/or seasonality in all seven cities but failed to reproduce the instantaneous O_3 variability, e.g., peaks in September 2019, Shenzhen [see Fig. S7(b)]. Agreed with the discussion above, this suggested that the periodicity of O_3 played a influential role in O_3 forecast. Combined with results from Fig. 7 and Fig. S5, LSTM can be quite reliable in forecasting O_3 of a relatively short period of future time (from 8 to 72 h). However, forecasting O_3 of a long period of time could be problematic as LSTM cannot capture the instantaneous variations.

E. Future Studies

Identifying factors that are influential to the LESO framework performance and quantifying their impacts is critical to framework applications in practice and to extending LESO in other countries/regions. Therefore, we will assess the impacts of environmental parameters (e.g., solar radiation and precipitation) on the framework performance through statistical metrics like the 'feature importance" [41]. In addition, 90% of the atmospheric O_3 concentrates in the stratosphere [42], thereby the contribution of satellite O_3 columns to estimating surface concentrations may not be significant. Regarding this, we will inter-compare estimation results from multiple satellites and results from tropospheric/total columns.

Identifying LESO applications in other countries/regions will be helpful for global O_3 pollution control. For example, Europe and North America have achieved successes in reducing the concentrations of particular matters and nitrogen oxides, but the levels of surface O_3 still exceed the limits for public health [43], [44]. The effectiveness and validity of applying LESO in these regions remains unknown because local emission sources, meteorological conditions, and air quality observation stations in these regions are expected to be different from CNEMC. In the future, we will explore the feasibility of applying LESO in a broader area.

IV. Summary and Conclusion

In this study, we proposed a spatial and temporal surface O_3 estimation model (LESO) and analyzed its performance and influencing factors. Using satellite observations and reanalyzed meteorology, LESO (including a DF21 O_3 interpolation and an LSTM O_3 forecast sub-models) was validated against *in situ* measurements and a benchmarking model in three Chinese mainly O_3-polluted regions. Results of DF21 for two satellites (Aura/OMI and Sentinel-5p/TROPOMI) agreed well. According to the validations and comparison with previous studies, our findings suggest that LESO can provide relatively accurate O_3 estimates with relatively fewer input drivers. By exploring the limitations of estimating spatiotemporal O_3 with machine-learning methods, we recommend:

1) Apply LESO or other similar methods in areas with at least three training sites;
2) The distance between sites should be less than 30 km;
3) Preferably forecast city-level O_3 within three days;
4) Be cautious in forecasting O_3 at fine resolution (e.g., site-level) or in areas may be experience fast O_3 concentration changes.

As we also discussed the number of training sites and their distance to the test site, our findings are also helpful to the selection of future site locations. In regard to the LESO application, we generated two sets of daily surface O_3 products covering the whole of mainland China for TROPOMI (2019–2020) and OMI (2014–2020) for open access.

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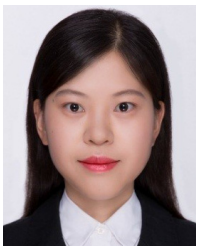
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